M.E. MODEL PROJECT REVIEW PACK

AVA Evaluation

An Empirical Study of Conflict Resolution for Assertional State in Multi-Agent AI Systems

Agree · Verify · Abstain



A provisional, classroom-ready review aid — not an official KEC template

STATUS: proposal/review only. Pass 3 literature verification is incomplete; novelty is unresolved; the convergence condition is a target, not a proven theorem. No protocol implementation or formal thesis/paper drafting is authorized yet.

At a glance

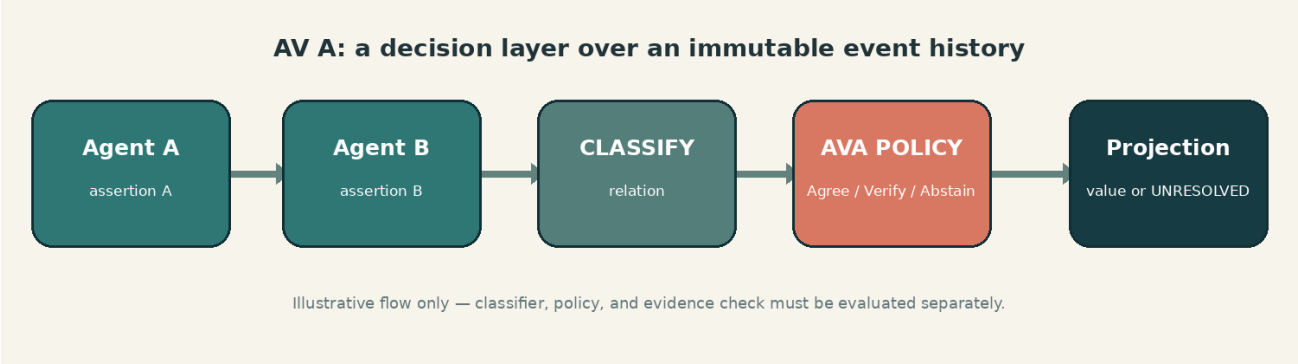
This pack turns the supplied finalized specification into a practical review presentation structure. The eight module titles below are provisional headings derived from that specification because no official college module list or review template was supplied.

LOCKED IN SPEC	PARTIAL LITERATURE EVIDENCE	OPEN / NOT AUTHORIZED

- Locked in the supplied spec: project name/title; same-key assertional-state focus; four relation labels; three policy actions; listed comparison arms; fault model boundaries; metrics and staged phases.
- Partial evidence: nearby studies already cover portions of semantic pair judgments, conflict-preserving state, provenance-aware resolution, abstention and asynchronous coordination. The comparisons below are scoped to the evidence saved so far.
- Open: exact novelty claim; WH-1/2/3 authoritative definitions; same-key schema and label rules; classifier/policy mapping; annotation capacity, sample size, budget and supervisor/college expectations.

60-second oral explanation

“When several AI agents update shared state, every replica can receive the same events while the meaning is still wrong or unsafe. AVA studies assertions about the same declared key: first classify the relationship, then decide whether to agree, verify with evidence, or preserve the disagreement as unresolved. I plan to compare deterministic, semantic, hybrid and structured methods using separate measures for false merges, unnecessary conflicts, calibration and cost, then test selected methods under delayed, duplicated and reordered delivery. SCP-MA is the test environment, not a claimed new protocol. The literature gap is still being checked, so this review presents a research proposal, not a confirmed novelty claim or completed implementation.”



Eight provisional review modules

1. Problem and background — Why shared meaning can fail even when replicas agree

2. Scope and state model — Separate additive facts from single-valued assertions

3. Relation classes — Classify the relationship before choosing a policy action

4. Agree / Verify / Abstain model — Keep policy actions distinct from terminal outcomes

5. Research questions and comparison arms — Compare strong baselines fairly, and separate classifier from policy

6. Dataset, annotation and metrics — Build a trustworthy gold set; report safety errors separately

7. SCP-MA experiment and convergence target — Use a minimal fault simulator only after the research gate

8. Risks, limitations and staged plan — Protect the research gate and keep the M.E. scope manageable

Provisional Module 1

Problem and background

Why shared meaning can fail even when replicas agree

Core idea

Several agents may write different statements about one shared fact. A replicated log can deliver every event to every replica and still leave the application with an unsafe or misleading meaning. AVA studies the decision made after same-key disagreement is detected; it is not a general consensus protocol or universal truth checker.

What to say / show

- Motivation: mechanical convergence is not semantic correctness.
- Question: which assertion relationships can be safely reconciled, and when should the system verify or abstain?
- Falsifiable stance: deterministic methods may beat semantic models; either result is valid.



Animal illustration is a teaching analogy only — not evidence or a model of the protocol.

Short talk track

"I am not claiming that AI must be better. I want to measure when each method can safely resolve a disagreement and when it should leave the state unresolved."

Likely review question

Ask: Is this a real systems problem? Answer: yes, as a bounded empirical problem; the size and originality of the gap still need verified literature comparison.

Provisional Module 2

Scope and state model

Separate additive facts from single-valued assertions

Core idea

The proposal separates state into additive state A and assertional state Q. Additive observations such as logs can be accumulated with standard CRDT structures. The focal case is a state key intended to have one value at a time, where competing values may conflict. An assertion records key, value, source agent, causal metadata such as a vector clock, and logical timestamp.

What to say / show

- In scope: pairs concerning the same declared key; cooperative agents; stale data; concurrent writes; delay, duplication, reordering, disconnection and reconnection.
- Out of scope: Byzantine agents, permanent partitions, general open-domain contradiction, cross-key intent conflicts and organizational governance.
- Key identity/cardinality and typed value schema must be specified before examples are labeled.

Short talk track

“The key is the anchor. If two values refer to different roles or time scopes, they may not be a conflict at all. The dataset must make that scope explicit.”

Likely review question

Ask: Why not solve every conflict? Answer: cross-key invalid combinations and adversarial behavior are separate research problems and are explicitly excluded.

Provisional Module 3

Relation classes

Classify the relationship before choosing a policy action

Core idea

The specification names four pair classes. Paraphrase means the same fact in different wording. Compatible means both statements can be true together. Soft conflict is a genuine difference that a stated policy or temporal context could resolve. Hard conflict means mutually exclusive assertions that are unsafe to auto-merge. The labels concern the pair's relationship, not the system's action.

What to say / show

- Paraphrase: "PostgreSQL is selected" / "The project uses PostgreSQL."
- Compatible: two statements can both hold under clearly declared distinct roles/scopes.
- Soft conflict: API v1 is in use while migration to v2 is underway, if the temporal transition is represented.
- Hard conflict: database=PostgreSQL versus database=DynamoDB for the same role and time.



Animal illustration is a teaching analogy only — not evidence or a model of the protocol.

Short talk track

"First I label what relation the pair has. Only after that do I ask whether a policy should agree, verify, or abstain."

Likely review question

Design risk: the provided PostgreSQL/DynamoDB compatible example describes different database roles, so it may violate same-key scope unless the key includes role/context.

Provisional Module 4

Agree / Verify / Abstain model

Keep policy actions distinct from terminal outcomes

Core idea

AVA has three policy actions: Agree, Verify, Abstain. Verify is intermediate. It consults additional evidence such as provenance, timestamps, causal metadata, or deterministic rules. If the evidence supports a safe resolution it terminates as verified-agree; otherwise it terminates in abstention. Abstention preserves the competing assertions as UNRESOLVED rather than deleting or guessing.

What to say / show

- Direct Agree: sufficient support for immediate reconciliation.
- Verify: do not merge yet; run an evidence check.
- Abstain: preserve conflict and defer resolution.
- In convergence-proof experiments, Verify must be a deterministic pure function of shared canonical inputs; a live stochastic model call is not convergence-safe.



Animal illustration is a teaching analogy only — not evidence or a model of the protocol.

Short talk track

"Verify is not just choosing the newest write. It is a second-stage evidence check. If evidence is not enough, the system keeps both claims unresolved."

Likely review question

Ask: How is Verify different from B1? Answer: B1 is a winner-selection rule; Verify asks whether additional deterministic evidence justifies a safe decision.

Provisional Module 5

Research questions and comparison arms

Compare strong baselines fairly, and separate classifier from policy

Core idea

The primary question is whether decision mechanisms can reliably classify same-key assertion pairs and determine when to agree, verify, or abstain. The specification lists seven arms: deterministic winner selection; embeddings; NLI; a hybrid SCS formulation; LLM-as-a-Judge; Jev (System-One decision model); and conflict-preserving abstention.

What to say / show

- RQ1–RQ2: relation classification and semantic-versus-deterministic performance.
- RQ3–RQ4: confidence-aware policy, calibration, latency and cost.
- RQ5–RQ6: asynchronous behavior and scaling. RQ7: trust-weighting ablation.
- Fairness issue: B1 and B7 are policies, while B2–B6 are primarily classifiers. Report classification separately from final resolution outcomes, with shared inputs and explicit mappings.

Short talk track

"I will not compare a classifier and a safety policy as if they do the same job. Classification quality and final resolution safety will be reported separately."

Likely review question

Open choice: determine whether the main thesis is a classifier benchmark, a policy comparison, or a narrow staged combination. Keep seven-arm breadth provisional until scope and resources are checked.

Provisional Module 6

Dataset, annotation and metrics

Build a trustworthy gold set; report safety errors separately

Core idea

Each record is intended to include assertion pair, declared key, agents, context, temporal context, evidence, gold relation label and difficulty. The proposed construction has controlled synthetic examples, LLM-generated realistic examples, and real traces only where lawful and practical. Ground truth requires multiple human annotators, agreement measurement and adjudication; an LLM alone is not a gold-label authority.

What to say / show

- Classification: accuracy, precision, recall, macro/per-class F1.
- Safety: FMR counts incorrect Agree outcomes on true conflicts; FCR counts failures to safely resolve true paraphrase/compatible cases. Unnecessary abstention is a named subset of FCR and is intentionally reported separately too.
- Calibration: ECE, Brier, reliability diagrams, risk–coverage, selective accuracy. Cost: latency, tokens/API calls, compute, throughput.
- Pre-register annotation guide, key schema, split strategy, leakage controls and sample-size rationale before benchmark implementation.

Short talk track

“A false merge can be dangerous, so I report it separately from needless abstention. I will not hide either inside one score.”

Likely review question

Open decisions: annotation capacity, sample size/power, adjudication protocol, domain balance and whether context/provenance is visible to each classifier.

Provisional Module 7

SCP-MA experiment and convergence target

Use a minimal fault simulator only after the research gate

Core idea

SCP-MA is the evaluation environment, not a claimed novel protocol. It models concurrent cooperative agents, immutable events, causal metadata, asynchronous delivery faults and eventual reconciliation. State is projected from an event history; semantic conflict does not trigger destructive rollback. The target property says that replicas with the same valid event set should compute the same deterministic projection after canonicalization.

What to say / show

- Validate event-set convergence in a semantics-free simulator before adding semantic decision methods.
- Inject delay, duplication, reordering, temporary disconnection and reconnection. Measure convergence rate/latency, divergence, messages/bytes, recovery and duplicate handling.
- For convergence-critical arms, classification/Verify must be deterministic or use decision caching/replay. Model-based Verify is a separate non-convergence-safe condition.
- Do not describe the target as a proved theorem: determinism, canonicalization and exact projected state remain obligations.



Animal illustration is a teaching analogy only — not evidence or a model of the protocol.

Short talk track

"The simulator tests whether delivery faults change the result. First I would prove the event mechanics with no semantic model; then add only methods that meet each arm's stated determinism conditions."

Likely review question

Ask: Is SCP-MA the novelty? Answer: no; it is the experimental scaffold. Any novelty claim must come from verified research question and evidence, not the simulator name.

Provisional Module 8

Risks, limitations and staged plan

Protect the research gate and keep the M.E. scope manageable

Core idea

The project remains at the literature and design gate. Pass 3 full-text/source-wide verification is incomplete, the novelty gap is unresolved, the exact WH-1/2/3 definitions are absent from the supplied spec, and convergence is a target rather than a proof. No protocol implementation or formal thesis/paper drafting is authorized yet. This pack is for review discussion only.

What to say / show

- Gate 1: finish primary-source and NLI/contradiction review; actively challenge the candidate gap.
- Gate 2: settle key schema, label rules, policy/classifier separation, evidence access, and safe Verify semantics.
- Gate 3: check college scope, annotation feasibility, compute/API budget and supervisor expectations; these facts are not supplied.
- If the question survives: prioritize a small reproducible benchmark and safety–utility analysis; keep the simulator minimal and scaling as stretch work. If not, narrow, reframe, or abandon rather than retrofitting novelty.

Short talk track

“My next step is not coding. It is to finish the literature attack and resolve the design decisions. If the gap does not survive, I will narrow or change direction instead of claiming novelty.”

Likely review question

Unverified facts not to invent: official KEC module names/template, supervisor requirements, deadlines, annotation pool, budget, results, and implementation status.

How the seven arms fit together

The supplied specification lists seven arms, but they do not all operate at the same layer. Treat the table as the planned comparison, not evidence that a fair design is already settled.

Arm	Role in the study	Fair-comparison note
B1 deterministic argmax(timestamp, agent_id)	Winner-selection baseline	Resolution policy; not itself a four-class classifier.
B2 embeddings	Semantic similarity signal	Similarity is not logical compatibility; evaluate empirically.
B3 NLI	Entailment / contradiction / neutral signal	Specify pair/context input consistently.
B4 hybrid SCS	Cosine-distance + distance term	Weights/formula are experimental choices, not assumed optimal.
B5 LLM-as-a-Judge	Generative pair judgment	Fix model/version/prompt/temp; repeat and report variance.
B6 Jev (System-One decision model)	Structured decision questions	Compare calibration, accuracy, cost with same evidence access.
B7 conflict-preserving abstention	Safety baseline: preserve as UNRESOLVED	Policy baseline; measure resolution coverage and safety separately.

Metrics in plain language

False-Merge Rate (FMR)

Among genuinely conflicting cases, the fraction incorrectly ended in Agree, whether direct or after Verify. Report this separately because an unsafe merge can be costly.

False-Conflict Rate (FCR)

Among genuine Paraphrase + Compatible cases, the fraction not safely resolved. The specification includes unnecessary abstentions here.

Unnecessary-abstention rate

The subset of FCR cases that reached Verify and then abstained despite resolvable ground truth. It overlaps FCR intentionally; these metrics are not additive.

Calibration and coverage

For probabilistic methods, check whether confidence matches observed correctness, and plot accuracy as the system abstains more or less.

Distributed behavior

Measure whether replicas converge to the defined projection, how long it takes, divergence during faults, message/byte costs, duplicate handling and recovery.

Literature: useful comparisons, carefully bounded

These are partial/qualified comparisons from the project's saved source audit, not a final systematic review. The aim is to show examiners that nearby work is being taken seriously, while avoiding a claim that the field has no prior work.

- MELD — A Protocol for Merging Knowledge Across Distributed Agentic Memories — Closest pair-taxonomy overlap; 104 balanced pairs deliberately use distinct claim keys; status convergence is not semantic-view convergence.
- StateFuse — Deterministic Conflict-Preserving Memory for Multi-Agent Systems — Conflict objects, provenance, immutable history, deterministic projection and abstention overlap; reviewed sections do not establish identical task.
- TOKI — A Bitemporal Operator Algebra for Contradiction Resolution in LLM-Agent Persistent Memory — Temporal records, provenance, typed operators, evidence/confirmation policies; section-level evidence in project notes.
- SubtleMemory — A Benchmark for Fine-Grained Relational Memory Discrimination — 1,522 downstream QA instances over 10 long histories; relational memory task, not isolated assertion-pair classification.
- S-Bus — Automatic Read-Set Reconstruction for Multi-Agent LLM State Coordination — Structural concurrency control; 45/45 OCC rejections in a specified experiment; not semantic classification.
- Semantic Consensus Framework (SCF) — Reported 27.9% precision, 65.2% recall, F1 39.0 for its workflow-intent conflict task; not same-key assertion pairs.
- CodeCRDT — Observation-Driven Coordination for Multi-Agent LLM Code Generation — 5–10% semantic-conflict estimate from manual inspection of 60/600 runs; preliminary, not general rate.
- Governed Shared Memory for Multi-Agent LLM Systems — Same-subject/property supersession study; 90/90 admitted conflicts vs 98/200 overall due to admission effects, as recorded in project notes.
- Reliable Post-Retrieval Assembly for Agent Memory — +10.8/+21 points are whole-pipeline result; executor-only comparison +2/0 and LongMemEval 26/45 vs 29/45, p=.45.
- ECon — On the Detection and Resolution of Evidence Conflicts — Question-conditioned evidence-pair conflict task; adjacent NLI baseline, not replicated keyed state.
- RefNLI — Reference ambiguity and context mismatch complicate contradiction labels.
- Selective Prediction for NLP — Selective prediction/abstention predates AVA; not direct distributed same-key experiment.

Numbers to state only with their scope

- MELD: the 104-pair balanced diagnostic set intentionally uses distinct claim keys; its reported pair-classification results therefore do not directly establish performance on a strict same-key-only dataset.
- CodeCRDT: the 5–10% semantic-conflict estimate is based on manual inspection of 60 of 600 runs, a preliminary sample rather than a validated general rate.
- SCF: reported Table 2 values are 27.9% precision, 65.2% recall and F1 39.0 for its workflow-intent conflict task; 65.2% is not accuracy.
- Deterministic post-retrieval assembly: +10.8 percentage points on average / +21 at context length 262K compares whole pipelines. The executor-only comparison is +2 average / 0 at 262K; on LongMemEval, 26/45 vs 29/45 with p=.45.
- SubtleMemory v3: 1,522 downstream QA instances across 10 persona-level histories and 1,090 relation-controlled variant sets. It tests long-history relational memory discrimination, not isolated assertion-pair classification.

Open decisions and review-ready answers

What exactly is the key?

Define cardinality, type, scope, role and temporal granularity. A compatible pair must either share one key by design or be clearly identified as a different-key/control case.

Are class labels independent of policy?

Yes in the proposed design: relation class is a gold annotation; Agree/Verify/Abstain is an action. Specify how each arm maps predictions/evidence to actions.

What does Verify inspect?

Specify which deterministic evidence is available and what threshold/rule is sufficient. For convergence experiments, both replicas with identical canonical evidence must produce identical Verify outcomes.

What is the exact thesis novelty?

Not established. The safest candidate distinction is a narrowly scoped empirical question about evidence-based Verify after same-key classification under asynchronous delivery, but StateFuse/TOKI/MELD and other work require full comparison.

What are WH-1, WH-2, WH-3?

The supplied finalized specification names but does not define them. Do not present any agent-written paraphrase as the owner’s exact hypotheses.

How large is the work?

Seven arms + a human-labeled dataset + calibration + a simulator + scaling is ambitious. Keep the core benchmark/safety results first; treat simulator depth and scaling as staged/stretch scope subject to requirements and feasibility.

What has been implemented?

Nothing in the thesis protocol is authorized or claimed implemented. The current work is planning and literature/design reconciliation.

Staged plan (no dates invented)

1. Stage A — Literature gate: source-wide/appendix verification, NLI/contradiction review, active attempt to disprove the candidate gap; record exact versions and review scope.
2. Stage B — Design preregistration: key schema/cardinality; label guide; evidence visibility; classifier/policy separation; split/leakage controls; sample-size and resource plan.
3. Stage C — Core benchmark, only if the gate and approval conditions are met: controlled dataset and human annotation, deterministic baseline, selected semantic methods, safety/calibration/cost reporting.
4. Stage D — Minimal distributed simulator: first test event canonicalization and deterministic projection without semantic methods; then integrate only conditions compatible with the convergence claim.
5. Stage E — Scope decision and writing: expand, narrow or reframe based on measured results and college/supervisor requirements; only proceed to implementation or formal thesis drafting after the project’s gate is satisfied.

Sources and status note

Primary source for the locked project choices: “AVA Evaluation — Finalized Thesis Specification” supplied by Selvin. Source links above point to public versions where available; literature summaries and metrics are quoted only with the scope limitations preserved in the saved project audit. Please verify versions/citations against full texts before formal submission.

This review pack is not an official college format, an approved thesis, a completed literature review, an implementation plan authorization, or a report of experimental results. It is an explanatory proposal aid made from the supplied specification and qualified evidence in conversation history.

Short closing pitch

“AVA asks a practical question: when agents disagree about the same state key, what can be reconciled safely, what needs evidence, and what should remain unresolved? I will compare methods with explicit safety and usefulness measures, without assuming that more intelligent models perform better. The current contribution is a testable proposal; the novelty and convergence claims stay open until their gates are met.”